CS 470 Final Reflection

Shawn Carrier

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https://youtu.be/-7EadXnYniM

Experience and Strengths

1. What skills have you learned, developed, or mastered in this course to help you become a more marketable candidate in your career field?

One thing I've learned from this degree program, and this course, is that it is difficult to master a language, skill, or toolset. However, over the eight weeks of this course I have learned how to create and manage containers, as well as how to use tools like Docker and Amazon Web Services to help manage containers, databases, and front-end-back-end relationships. Setting up access control policies and roles gave further insight into the rule of least access when it comes to users and clients.

2. Describe your strengths as a software developer.

My greatest strength is my ability to analyze and interpret a problem. I can take a problem, think about all the pieces to it and goals to achieve, and break them down into smaller parts to streamline the development process, ease of access for other team developers, and use a documentation style that is neither too wordy nor succinct. Having been able to work with multiple languages over the course of the degree program has helped me interpret other languages to spend as little time as possible learning local nomenclature.

3. Identify the types of roles you are prepared to assume in a new job.

There are various roles that I would feel comfortable taking on in a new position. Back-end development was the focus of my degree, but I would also be comfortable with front end development. A position where I can work with the code or logic behind the code and systems are roles I feel prepared to perform. Unrelated, a role in IT would be a nice fit with a background in programming.

Planning for Growth

1. Identify various ways that microservices or serverless may be used to produce efficiencies of management and scale in your web application in the future.

Splitting applications into smaller sub-applications would be able to produce better efficiency and ease development on specific parts without touching everything that is the application. Integration of MongoDB, while potentially incurring cost, would offer easier ways to filter databases compared to what DynamoDB offers. Splitting APIs set up for a full stack application into smaller, singular ones would allow for quicker calls and less vectors of attack regarding cybersecurity. When an application becomes too large it becomes harder to trace vulnerabilities, understand the scale of dependencies, and trace requests across the environment.

2. Explain several pros and cons that would be deciding factors in plans for expansion.

Regarding AWS, APIs through AWS Lambda work on a pay-as-you-go service. Using containers is cheap to set up initially, but accrues cost when running 24/7, even if they aren't being used. Using a local server is nice for complete control over your application, but using a serverless system affords near constant uptime in times of maintenance and could cost a lot less long term. Serverless computing also offers faster development to deployment times than non-serverless computing.

3. What roles do elasticity and pay-for-service play in decision making for planned future growth?

Serverless computing is a volatile system, in that it is hard to predict the popularity or use cases for your application. This leads to issues in scaling and, potentially, errors. Having the ability to split your application into smaller components can help mitigate errors, but using cloud-based serverless solutions can aid with the elasticity of your application. With most options for serverless computing incorporating a pay-as-you-go model, you can, quite literally, pay for what you need as you use it instead of paying in advance for something that isn't being used while maintaining complete uptime. Containers are nice for deployment and can be used in conjunction with a service like AWS and their pay-as-you-go model for serverless computing.